

When can a simulator-trained screen be trusted? A reliability map for machine-learning triage of Ariel transmission spectra

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Abstract

Machine-learning screens have been proposed to decide which of the ~1,000 exoplanets Ariel will observe deserve deeper follow-up, including a Tier-1 classification strategy published by the Ariel consortium. Because no Ariel data exist, every such screen has been trained and validated on spectra from the same simulator, so its reported accuracy does not measure its behaviour on atmospheres whose physics differs from the simulator's. This note presents a workflow for mapping when such a screen can be trusted, built from standard tools, and demonstrates it on two screens: the consortium's Tier-1 molecular screen, re-implemented from its published description because its code was not released, and a screen built from scratch for this note for a stated Ariel science target, the carbon-rich ($C/O > 1$) regime. The simulator was broken on purpose along ten physically motivated axes; for each, the workflow measured the loss, the part no retraining could recover, what a randomized training grid absorbed, whether a decline rule flagged the errors, and what removed the loss, at the Tier-1 binning where triage would occur and on Ariel's actual target list. The map had three regions. Modelled mismatch was largely absorbed and the residual was irreducible information loss. Disequilibrium chemistry helped. Physics the simulator omitted could not be trusted: HCN and C_2H_2 , present at $\sim 10^{-5}$ in carbon-rich atmospheres and absent from the field's standard Ariel training grid, dropped the carbon-rich screen from 96.5 % to 71.9 % — chance on carbon-rich planets, with confident probabilities; no retraining on other ingredients helped, only a distance-based novelty score detected it, and adding the two species restored 96.4 %. The same omission left the molecule-presence screen unchanged: omitted physics was fatal when it sat on the feature that defined the class. At Tier 1, randomization still helped but no decline rule worked and most of every serious loss was unrecoverable by any training. The workflow is released as scripts that take any screen and any forward model.

Keywords: exoplanet atmospheres; transmission spectroscopy; Ariel; machine learning; classification; domain shift; simulation-based inference; out-of-distribution detection; selective prediction; forward-model misspecification; carbon-to-oxygen ratio; disequilibrium chemistry; HCN; acetylene; stellar contamination; opacity databases

1. Introduction

Ariel, launching in 2029, will obtain transmission spectra of roughly a thousand transiting exoplanets between 0.5 and 7.8 μm (1). The survey is tiered: a reconnaissance pass over all targets at coarse binning (Tier 1), a chemical survey of a few hundred at higher binning (Tier 2), and repeated observations of a few dozen at the instruments' native resolution (Tier 3). A planet reaches a tier when its spectrum, binned as that tier prescribes, is expected to reach a signal-to-noise ratio of seven on an assumed atmospheric modulation after a number of transits computed with the mission's radiometric model (2-4); the prescribed binnings are approximately $R \approx 1, 3$ and 1 in the NIRSpec, AIRS-CH0 and AIRS-CH1 channels at Tier 1 and $R \approx 10, 50$ and 10 at Tier 2 (5). Which planets advance from the reconnaissance pass to the deeper tiers is a decision that has to be made for a thousand objects.

Machine learning has been proposed for that decision. The Ariel consortium's population study of the Tier-1 survey presented "a strategy to select candidate planets for reobservation in Ariel's higher resolution Tier", a band metric to classify planets by composition without a retrieval, and four scikit-learn classifiers trained on simulated Tier-1 spectra to flag the presence of individual molecules, reaching 64–89 % per molecule (6). The annual Ariel Data Challenges have drawn some 23,000 submissions to atmospheric-inference tasks on a database of 105,887 simulated spectra (7-9), and a 2026 review counted machine-learning retrieval, detrending and surrogate modelling among the methods being prepared for the mission (10). Every one of these was trained on a forward model of an atmosphere and a model of the telescope, because no spectrum of Ariel quality exists for any target; the same is true of the machine-learning retrieval lineage from random forests to neural posterior estimation and flow matching (12-15).

Each was also validated the same way: on spectra held back from the same simulator. That test gives a reassuring number, but training and testing share every assumption the simulator makes — which molecules exist, which database provides their absorption, whether there are clouds or haze, how the host star's spots contaminate the signal, how the noise is distributed — and the real atmospheres will differ from those assumptions in ways the test cannot expose. The consortium's Tier-1 study said so itself: its spectra "will only be used as 'transmission spectral shapes' to test our methods against", with no claim about the atmospheric model's realism (6). The nearest exception in the literature tested a retrieval network on spectra with an absorber added, an absorber removed, and unocculted star spots, and found it more robust than nested sampling on those three changes (11); it did not ask what could be done about the losses, whether the network could tell, or how the answer depended on the binning at which a decision is made. Classical retrievals have had their forward-model dependence priced — across codes (39), across the equilibrium assumption (20) — and misspecification of simulation-trained inference is an active concern in cosmology (37, 38); the question had not been asked of exoplanet screens, where the mismatch is physics rather than simulation fidelity, and the one Ariel dataset built with a train/test shift varies the instrument, not the atmosphere (45).

A screen that fails quietly returns confident answers, and a mission acts on them. This note takes the question a mission faces before launch — *when can a simulator-trained screen be trusted, and when can it not?* — and turns it into a workflow that can be run on any screen: on data whose physics differs from the training simulator, in ways one anticipates and in ways one deliberately holds out, which planets does the screen still classify at a stated accuracy, does it know which planets it cannot, and what fixes what? The workflow was assembled from standard machine-learning tools and applied to two screens: the consortium's published Tier-1 screen, re-implemented from its paper because its code was not released, and a second screen built from scratch for this note. The findings reported are what that application produced.

1.1 Contributions

The workflow (Section 2.4) combines five standard measurements — loss under a re-rendered mismatch, the ceiling from a screen trained at the test condition, absorption by a randomized training grid with each ingredient held out in turn, decline rules judged against the clean selective baseline, and evaluation at the decision tier and on the mission's target list — into one reliability map, and is released as scripts that take any screen and any forward model. No method in it is new: domain randomization, selective prediction, distance-based novelty scores, oracle bounds and held-out-axis evaluation are standard in machine learning and are cited where used; the assembly, the physics-side mismatch axes, and the absorb-or-detect-or-fix reading of each row are what the workflow adds.

Applied to the two screens — the consortium's Tier-1 design, re-implemented from its published description, and the screen built for this note for a stated Ariel science target — the workflow produced three results. First, the map itself, for ten simulator mismatches at Tier 3 (Section 3.1). Second, that two molecules chemistry has long assigned to carbon-rich atmospheres, and that the field's standard Ariel training grid omits, put a carbon-rich screen at chance on exactly the planets it exists to find, with high confidence, while leaving a molecule-presence screen unchanged (Section 3.2). Third, the map at Tier 1, where triage would actually happen, where it differed materially from Tier 3 (Section 3.3).

2. Materials and Methods

2.1 *The consortium's Tier-1 screen, re-implemented*

The screen of reference (6) was re-implemented from the text as literally as it allowed; neither its simulator wrapper (Alfnoor) nor its trained classifiers were released. Its training population (POP-III) took the Ariel candidate list, repeated each planet four times, drew the atmospheric temperature uniformly in 0.7–1.05 of the planet's equilibrium temperature, the CH₄, H₂O, CO₂ and NH₃ abundances log-uniformly in 10⁻⁹–10⁻², and a grey cloud deck log-uniformly in 5 × 10²–10⁶ Pa, in an H₂/He atmosphere; its test population (POP-I) used each planet once with abundances in 10⁻⁷–10⁻². Spectra were binned to the seven Tier-1 points (three photometric bands, one NIRSpc point, two AIRS-CH0 points, one AIRS-CH1 point) and scattered with the Tier-1 requirement noise. Each molecule was labelled present if its abundance exceeded 10⁻⁵, 10⁻⁴ or 10⁻³; inputs were normalised to zero mean and unit variance; the four classifiers were scikit-learn's k-nearest neighbours, multilayer perceptron, random forest and support-vector classifier at default settings (47). The re-implementation used the 965 known planets of the current Mission Candidate Sample (3) as the planet list, the forward model of Section 2.2 in place of theirs, and the consortium simulator's noise shape (Section 2.3). At the 10⁻⁴ threshold it reached CH₄ 70–72 %, H₂O 63–69 %, CO₂ 60 % and NH₃ 76–78 % against the published 82–87, 71–78, 79–83 and 82–87 %. The shortfall is most likely on the re-implementation's side: CH₄, H₂O and NH₃ reached the published values when its noise was reduced to about a third, which points to the per-target noise (theirs came from ArielRad after the integer number of transits each target needs; the re-implementation's was set exactly at the requirement), while CO₂ did not reproduce at any noise level or with alternative placements of the AIRS-CH0 split, which points to the binning of its 4.3 μm band or to the opacity tables. The deviations and the checks are listed in Appendix A. Where the re-implementation fell short of the published Table 6, the mismatch costs were reported as differences from its own clean values.

2.2 *A carbon-rich screen, built from scratch*

The second screen was built from scratch for this note and targeted a chemical regime rather than a molecule: is the atmospheric carbon-to-oxygen ratio above one? The C/O ratio is a stated Ariel science objective, a tracer of where and how a planet formed (1), and C/O = 1 is a physical boundary — above it, oxygen is locked in CO and water is depleted, below it carbon is (16, 17) — so the label is a property of the atmosphere rather than a threshold imposed on the labelling convention. The grid held 18,040 training and 9,016 test planets in five independent test sets (18,156 and 9,072 were drawn; 0.4–0.9 % failed to render) with bulk parameters drawn independently over ranges informed by the Ariel target samples: planet radius 1–26 R_⊕, mass 1–300 M_⊕, host temperature 2,500–7,500 K, atmospheric temperature 500–2,500 K, C/O uniform in 0.2–1.8 and metallicity uniform in –1 to 1.5 dex. Abundances of H₂O, CH₄, CO, CO₂, NH₃ and O₃ followed from chemical equilibrium computed with FastChem (19) at the photospheric level (10⁻² bar), held constant with altitude, and the label

was $C/O > 1$, which the draw balanced to 0.48–0.52 on every split. Spectra were rendered with MultiREx over TauREx 3 (23, 24) at native resolution, without noise, and binned by exact integration to the Ariel Tier-3 layout: three photometric bands below $1.1\ \mu\text{m}$, then $R = 15$ over $1.10\text{--}1.95\ \mu\text{m}$, $R = 100$ over $1.95\text{--}3.90\ \mu\text{m}$ and $R = 30$ over $3.90\text{--}7.80\ \mu\text{m}$, 102 bins in all (2-4). Opacities were Exo-Transmit's tables (25), derived from the compilation of reference (42) with the line-list sources of reference (43); tables for CO, NH_3 , HCN and C_2H_2 were added from the same source. Noise was Gaussian with a per-spectrum standard deviation equal to the peak-to-peak amplitude divided by 15, shaped in wavelength by the noise-to-signal curve of an ExoRad model of the payload for the host's temperature (4); the shape of the consortium's ExoSim 2 simulator (27) agreed with it to Spearman 0.985 for solar-type hosts and 0.80 for the coolest, and replacing one shape with the other cost the screen 0.1 points.

Nine pipelines were tuned by cross-validation on the training set: XGBoost (46), random forests and multilayer perceptrons on raw bins, on principal components, and on per-spectrum-normalised bins. Per-spectrum normalisation (subtracting each spectrum's mean and dividing by its standard deviation, so the classifier sees shape rather than depth) was worth four to ten points over the alternatives, and normalised XGBoost was the screen used throughout: $96.47 \pm 0.26\%$ across the five test sets, $96.32 \pm 0.08\%$ across ten training draws, expected calibration error 0.009. Moving the label's cut away from $C/O = 1$ and retraining lowered accuracy to 94.8 % at $C/O = 0.79$ and 85.7 % at 1.26 while the majority-class rate rose to 63 and 66 %, so what the screen learned was the chemistry transition at $C/O \approx 1$, not an arbitrary line.

2.3 The mismatch axes

Each axis re-rendered the *same* test planets with one ingredient changed, so every loss was a within-planet difference. The axes and the concern each represents were: an optically thick grey cloud deck at 10^5 to $10\ \text{Pa}$ (aerosols hide the lower atmosphere); a Lee et al. Mie haze at 2×10^5 to $10^{10}\ \text{m}^{-3}$ (22); unocculted star spots at 2–20 % coverage, with the transit-light-source correction built from PHOENIX spectra at a spot contrast of 0.85 (21); white and time-correlated noise at effective signal-to-noise 12 to 5; a gain ramp across the detector; the same planets rendered by an independent radiative-transfer code, Exo-Transmit (25), with the same opacities; the same planets rendered with the ExoMol cross sections (26) in place of Exo-Transmit's tables for H_2O , CH_4 , CO_2 and CO; quenched chemistry, in which the carbon–oxygen partition was frozen at the level where the mixing timescale for an eddy-diffusion coefficient K_{zz} overtook the CO– CH_4 conversion timescale of reference (18) on a Guillot profile anchored to the isothermal temperature, for K_{zz} from 10^7 to $10^{11}\ \text{cm}^2\ \text{s}^{-1}$; HCN and C_2H_2 added at their FastChem equilibrium or quenched K_{zz} abundances (Section 3.2); and compounds of spots, haze and noise on the same planet. Cloud decks at 10^2 and $10\ \text{Pa}$, haze at $10^{10}\ \text{m}^{-3}$, faculae and the two non-native codes lay outside anything the randomized training grid contained and were tagged as such. Table 1 lists the axes, the concern each represents, and whether the randomized training grid of Section 2.4 contained it.

Table 1. The mismatch axes. Each re-rendered the same test planets with one ingredient changed; the last column says whether the ingredient was drawn at random in the randomized training grid, so that a test on it was in-range, or was never shown to the screen, so that a test on it was a held-out axis.

Axis	What is varied	Concern it represents	In the randomized grid
Cloud deck	grey, optically thick cloud top at 10^5 , 10^4 , 10^3 , 10^2 , $10\ \text{Pa}$	aerosols hide the lower atmosphere	$10^3\text{--}10^5\ \text{Pa}$, 50 % of planets

Axis	What is varied	Concern it represents	In the randomized grid
Haze	Lee et al. Mie haze, 0.1 μm , at 2×10^5 , 2×10^6 , 3×10^7 , 2.4×10^8 , 10^{10} m^{-3}	photochemical haze mutes and slopes the spectrum	$10^5\text{--}3 \times 10^8 \text{ m}^{-3}$, 60 %
Stellar contamination	unocculted spots at 2, 5, 10, 20 % coverage (contrast 0.85); faculae 5, 10 %; mixed	the transit light-source effect imprints the star on the planet	spots 0–20 %, 70 %
Noise level and colour	white and time-correlated ($\sigma = 3$ bins) at effective SNR 12, 10, 8, 5; a gain ramp $\times 0.25\text{--}2$	the deployed noise differs from the training noise	SNR 5–15, all planets
Radiative-transfer code	the same planets rendered by Exo-Transmit with the same opacities	code-to-code differences in the forward model	never (held out)
Opacity database	ExoMol cross sections for H_2O , CH_4 , CO_2 , CO in place of Exo-Transmit's tables	line-list differences between databases	never (held out)
Quenched chemistry	carbon–oxygen partition frozen at the quench level, $K_{\text{zz}} = 10^7\text{--}10^{11} \text{ cm}^2 \text{ s}^{-1}$	vertical mixing drives the photosphere out of equilibrium	50 % of planets
Omitted absorbers	HCN and C_2H_2 added at their FastChem equilibrium or quenched abundances	species the training forward model does not contain	never (held out)
Compounds	spots 10–20 % $\times$ haze $2 \times 10^6\text{--}3 \times 10^7 \times \text{SNR}$ 8–10 on the same planet	real planets are off on several axes at once	jointly, through the draws above
Binning and targets	Tier-1 (7 points) and Tier-2 (51) layouts; the 965 known Mission Candidate Sample planets under the mission's noise definition	the decision is made at Tier 1, on real targets	—

2.4 The measurements

Five quantities were measured for every axis, all on identical test planets and noise realisations, and each relative quantity names its reference. (i) *Loss*: the accuracy the clean-trained screen gave up. (ii) *Irreducible loss*: clean accuracy minus the accuracy of an *oracle* screen trained only at the test condition — the target-only bound of domain adaptation — so that any recovery was measured against what retraining could achieve rather than against clean accuracy. (iii) *Absorption*: the accuracy of a screen trained on a *randomized* grid (28) (noise augmentation acts as a regulariser and cannot undo a particular draw (29); augmentation with a transformation teaches invariance to it (40)) in which every planet carried a random haze density (log-uniform, $10^5\text{--}3 \times 10^8 \text{ m}^{-3}$, 60 % of planets), cloud-top pressure ($10^3\text{--}10^5 \text{ Pa}$, 50 %), spot coverage (0–20 %, 70 %), noise level (SNR 5–15) and equilibrium-or-quenched chemistry, drawn jointly; the same grid rebuilt without one ingredient gave the *held-out* variant that tested whether robustness transferred to physics never seen, the held-out-corruption protocol of reference (41). (iv) *Detection*: five decline rules — the probability margin (30), the disagreement of an XGBoost/random-forest/perceptron ensemble (33), the Mahalanobis distance from the training cloud in feature space (31), the k-nearest-neighbour distance (32) and a PCA reconstruction error — each thresholded to decline 10 % of *clean* test planets and never tuned on a shifted set; a rule was credited only with the accuracy of the accepted planets *above* the clean selective baseline at the same coverage (34), because declining low-confidence planets raises accuracy on clean data too. (v) *Fix*: what removed the loss — training on it, observing longer, or changing the forward model. The whole set was computed at Tier 3 and

again at Tier 1 (seven points, Section 3.3). Expectations for the randomized grid, the omitted species and the consortium screen were committed to the repository before the corresponding runs (Appendix E); the misses are reported as misses.

2.5 Assumptions and their expected effects

The workflow rests on modelling choices that the results depend on; each is stated here with its justification and, where it was measured, its effect. *Parameters drawn independently*: the grid has no main sequence and no correlation between planet and host; this keeps the sensitivity measurements clean but produces planets that do not exist. Its effect was measured on the mission's real target list (Section 3.3): 57 % of the known targets fell inside the grid's bulk-parameter box, and those outside scored no worse than those inside. *Chemical equilibrium at one pressure, held constant with altitude*: the label is exact for this chemistry, and the disequilibrium the assumption ignores was priced as an axis (quenching, with the eddy-diffusion coefficient swept over four orders of magnitude; the result did not depend on it); photochemistry was not priced. *An isothermal atmosphere*: the temperature–pressure structure was varied only through the quench profile; three-dimensional effects were not priced. *Noise proportional to each planet's amplitude*: this makes faint planets as easy as bright ones; its effect was bounded by an absolute-noise axis (20–200 ppm cost 3.3–10.8 points) and by the target-list test, where the noise was set instead by the mission's own tier requirement and the achieved signal-to-noise on the planets' real amplitudes was 15.6 at the median, close to the grid's convention. *Spot contrast fixed at 0.85*: contrasts of 0.80 and 0.90 moved the 20 %-coverage accuracy by -1.0 and $+1.8$ points. *Correlated-noise kernel of three bins*: kernels of one, three and eight bins cost 5.9, 7.3 and 5.9 points at SNR 8, against 4.1 for white noise. *One training draw*: ten resamples of the training set changed the clean accuracy by ± 0.08 . *The label's cut at $C/O = 1$* : justified by the chemistry and by the cut-sensitivity test of Section 2.2. *The consortium screen re-implemented from its text*: the reproduction is partial, and the reasons are discussed in Section 2.1 and Appendix A. *The mismatch axes themselves*: every mismatch was generated by the same family of simulators, so the map is conditional on the axis set; holding axes out narrows this and real spectra would test it, but nothing removes it (Section 3.5).

3. Results and Discussion

3.1 The map at Tier 3

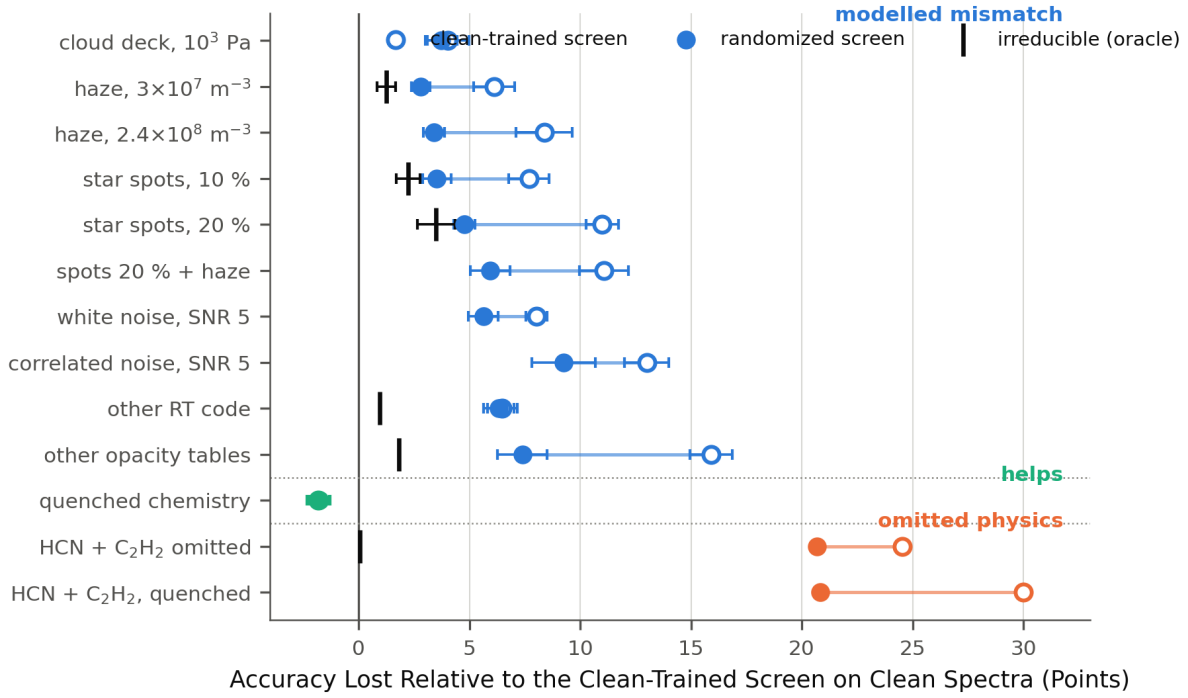


Figure 1. The reliability map at Tier 3. Each row is one mismatch; the open circle is the accuracy lost by the clean-trained screen, the filled circle the loss of the randomized screen, and the black tick the irreducible part — clean accuracy minus a screen trained only at the test condition. Rows are grouped by region: modelled mismatch (blue), mismatch that helps (green) and physics the simulator omitted (orange). Error bars are the standard deviation across the five test sets. The clean-trained screen scored 96.47 % on clean spectra; the randomized screen 95.41 %.

Table 2 and Figure 1 give the map at Tier 3, and it had three regions.

Table 2. The reliability map at Tier 3, one row per mismatch. Loss is in accuracy points relative to the clean-trained screen's 96.47 % on clean spectra; irreducible is clean accuracy minus the oracle trained at the test condition; randomized is the loss of the screen trained on the randomized grid (95.41 % clean); detect names the decline rule whose score ranked the screen's errors with AUROC ≥ 0.85 ; fix is what removed the loss.

Mismatch	Loss	Irreducible	Randomized	Detect	Fix
Cloud deck, 10^3 Pa	4.0	—	3.7	confidence	train on it
Haze, 3×10^7 m $^{-3}$	6.1	1.3	2.8	confidence	train on it (95 % of ceiling)
Star spots, 20 %	11.0	3.5	4.8	confidence	train on it (100 %); M-dwarf hosts lost 16, FGK 9
Spots 20 % + haze	11.1	—	5.9	confidence	sub-additive; randomization helped
White noise, SNR 5	7.5	4.8	5.6	confidence	mostly irreducible; observe longer
Correlated noise, SNR 5	12.3	4.6	9.3	confidence	idem; in-range training 86 % of ceiling
Other radiative-transfer code	6.5	1.0	6.3	confidence	only training on that code (85 % reducible)

Mismatch	Loss	Irreducible	Randomized	Detect	Fix
Other opacity tables	15.9	1.8	7.4	confidence and distance	training on them (88 % reducible)
Quenched chemistry	-1.8 (gain)	—	-1.8	nothing to detect	none; held for K_{zz} 10^7 – 10^{11}
HCN + C ₂ H ₂ omitted	24.5	0.0	20.7	distance only	add the species: 96.4 %
HCN + C ₂ H ₂ , quenched	30.0	—	20.8	distance only	idem

Region 1: mismatch that was modelled. Clouds, haze, star spots, noise, a gain ramp, another radiative-transfer code and another opacity database each cost the clean-trained screen 4 to 16 points at the strengths shown. Training on any one of them, at the strength deployed, recovered 95–100 % of what the oracle reached for spots and haze and 84–107 % for noise when the training noise levels included the test level; the residual was the irreducible part, 0.3–3.5 points for the deterministic re-renders and 4.6–4.8 points for noise at SNR 5, where the information had been removed from the data and only observing longer would restore it. The randomized grid — one screen trained with every ingredient drawn at random — halved the loss on clouds, haze, spots and compounds at a cost of 1.06 points on clean spectra, but reached only 31–83 % of the single-axis ceilings; the pre-registered expectation of ≥ 90 % was wrong, and joint randomization had a capacity price that single-axis training did not. Robustness transferred unevenly to ingredients the randomized grid never contained: 23 % of the achievable improvement for spots, 40 % for aerosols, 49 % for noise; the loss under the other opacity database fell from 15.9 to 7.4 points without those tables ever being seen, while the loss under the other radiative-transfer code did not move. Compound mismatches were sub-additive on this screen (spots and haze together cost 6 points less than the sum of their parts), so compounding was not the danger; the single largest modelled risk was the opacity database, 88 % of whose cost was reducible only by training on the alternative. Stellar contamination was the one axis whose cost depended on the host star: at 20 % coverage M-dwarf hosts lost 16 points and F, G and K hosts 9, as expected from the transit-light-source effect (21); every other axis was flat across host type.

The decline rules behaved the same way throughout this region: the probability margin and ensemble disagreement ranked the screen's errors well (AUROC 0.80–0.92) and the distance scores did not (0.37–0.72, mostly near 0.45), so confidence was the right instrument for a distortion of known physics. But no rule restored clean-level reliability. With the ensemble rule fixed to decline 10 % of clean planets, every in-range mismatch kept its accepted planets at 94.9 % or better, at coverage that fell to 65 % at SNR 5 — and the credit against the clean selective baseline at the same coverage was negative for every rule on every distorted axis (mean -4 points): declining bought back part of the loss, never all of it. A split-conformal calibration (36) that guaranteed 90 % coverage on clean data delivered 63–88 % under these mismatches — the guarantee is valid for the calibration distribution and needs the deployment distribution to hold (44) — and its prediction sets came out empty rather than ambiguous, because the screen was wrong with confidence (Appendix C).

The absorb/detect trade-off. Randomizing an ingredient into training made spectra carrying it look in-distribution, so the alarm for that ingredient went off: the Mahalanobis shift score's ability to separate shifted from clean spectra fell from 0.84 to 0.51 for 20 % spots, 0.91 to 0.64 for haze and 1.00 to 0.35 for the compound, while ingredients the grid never contained stayed fully visible (opacity tables 0.98, noise 1.00). The margin's error ranking, meanwhile, improved under randomization (0.85 to 0.91 for spots). A screen could not be both robust to a mismatch and able to warn about it; a design that wants both keeps a separate, deliberately

fragile model as the alarm.

Region 2: mismatch that helped. Quenching the carbon–oxygen partition at depth *raised* the clean-trained screen's accuracy from 96.5 to 98.3 %, by 6.6 points on planets below 1,000 K, at every eddy-diffusion coefficient from 10^7 to 10^{11} $\text{cm}^2 \text{s}^{-1}$. The reason is chemical: at depth a carbon-rich atmosphere is depleted of water by orders of magnitude (16, 17), quenching carries that depletion up to the photosphere, and the label's separation in water abundance on cool planets grew from 0.3 to 3.3 dex. The reverse deployment — a screen trained on quenched atmospheres applied to equilibrium ones — lost 10.1 points, of which mixing recovered 72 %. The design rule that follows is to train on the chemistry in which the label is hardest to see. Whether this holds under photochemistry, which the forward model does not include, was not tested.

Region 3: physics the simulator omitted. The last two rows of Table 2 are the subject of Section 3.2.

3.2 The omitted species

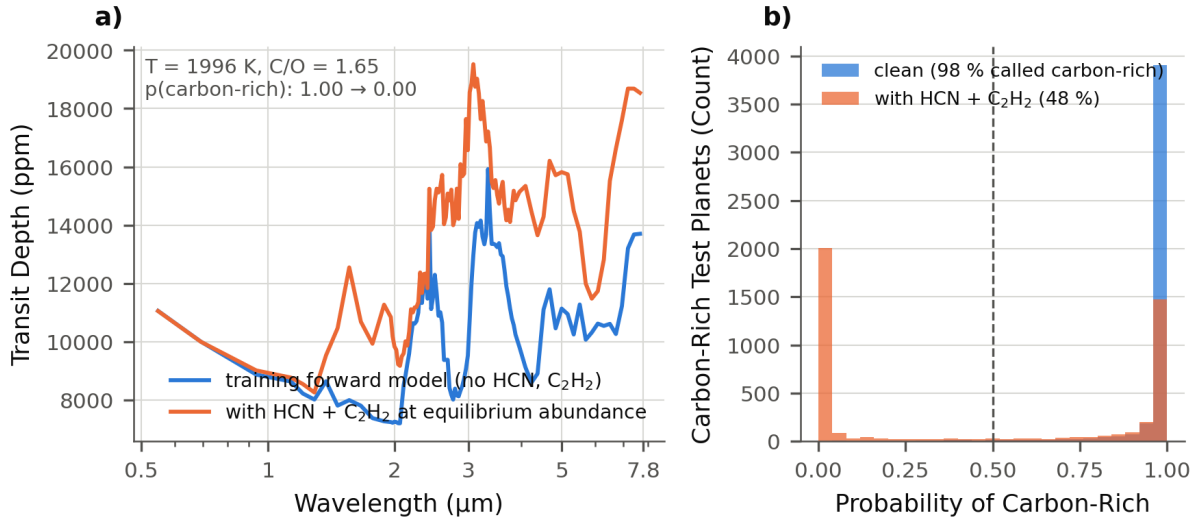


Figure 2. The omitted species. (a) A carbon-rich test planet at the Ariel Tier-3 binning, rendered by the training forward model (blue) and with HCN and C_2H_2 added at their equilibrium abundances (orange); the planet shown had the median effect among hot carbon-rich planets. The clean-trained screen's probability of "carbon-rich" for this planet fell from 1.00 to 0.00. (b) The same probability for all 4,508 carbon-rich test planets, clean (blue) and with the two species present (orange): the share called carbon-rich fell from 98 % to 48 %.

In an atmosphere with $\text{C/O} \geq 1$ above about 800 K, HCN and C_2H_2 become major constituents, enhanced by three to six orders of magnitude over their solar-composition values, and "can be considered good tracers of the C/O ratio" (16); disequilibrium chemistry enhances them further (17). FastChem gave them at $\sim 10^{-5}$ and $\sim 3 \times 10^{-5}$ on the carbon-rich planets of the grid (5×10^{-5} and 1.5×10^{-4} when quenched) and at 10^{-10} and 10^{-14} on the others. Neither molecule was in the forward model that trained the screen. Neither is in the Ariel Data Challenge database, whose forward model states that "the trace gases are H_2O , CH_4 , CO , CO_2 and NH_3 " (7), nor in the challenge that followed it (8), nor in the free-chemistry training models of reference (11). The omission is the field's standard, not particular to this grid.

Re-rendering the test planets with the two species added at their own abundances, with everything else unchanged and the six training gases verified to be identical, dropped the clean-trained screen from 96.47 % to 71.93 % (71.9 ± 1.1 across the five test sets). Carbon-rich planets went from 97.6 % to 47.6 % — chance — while planets with negligible HCN and C_2H_2 were unaffected (96.2 %), which was the built-in null control; the share of test planets called

carbon-rich halved from 0.52 to 0.26, so the screen called carbon-rich planets oxygen-rich. With the species at their quenched abundances the screen fell to 66.5 %, 36 % on carbon-rich planets. The loss was not the property of one model: normalised random forests and perceptrons lost 23 points, XGBoost on principal components 14 and on raw bins 16; the Tier-2 screen fell from 96.0 to 70.7 % and the Tier-1 screen from 88.7 to 66.3 %, with carbon-rich planets at 45–46 % in both.

The mechanism was only partly the band overlap one would guess. On a hot carbon-rich planet of 1,400 ppm feature amplitude, HCN at 10^{-5} added 300–600 ppm of absorption in bands at 1.5, 2.0, 3.0, 4.8 and 7 μm and C_2H_2 at 3×10^{-5} added 200–470 ppm at 1.5, 3.0 and 7 μm — a third of the planet's own signal, spread across the range (Figure 2a). Restoring only the 2.75–3.05 μm bins, where the added absorption fills the gap between the water and methane bands, to their clean values recovered 5 of the 24.5 points; restoring 2.75–4.3 μm recovered 8. The rest was broadband: absorption at a third of the amplitude across 1.5–7 μm moved every bin once the spectrum was normalised, which is why the normalised pipelines lost the most.

What could be done about it is the point (Figure 3). The randomized grid, with every other ingredient varied, did not help: 75.8 %. The screen was confidently wrong: its expected calibration error rose from 0.009 to 0.25, the mean probability it assigned to a true carbon-rich planet fell from 0.96 to 0.46, the conformal coverage fell from 91 to 64 % with the sets empty, and the probability margin ranked the errors at AUROC 0.66 (Figure 2b). The distance scores, useless on every distorted axis, ranked these errors at 0.90–0.92 and kept 95 % accuracy on the 61 % of planets they accepted; on the randomized screen they were the only rules anywhere in the envelope whose credit against the clean selective baseline was positive (+0.3 to +4.3 points). A spectrum with a new absorber is *novel*, so distance from the training set was error; a distorted spectrum of known physics was not. And the loss was entirely reparable in the forward model: a screen trained with HCN and C_2H_2 in the grid scored 96.43 % on the same test set, an irreducible loss of -0.04 points, marginally above the original screen on its own data.

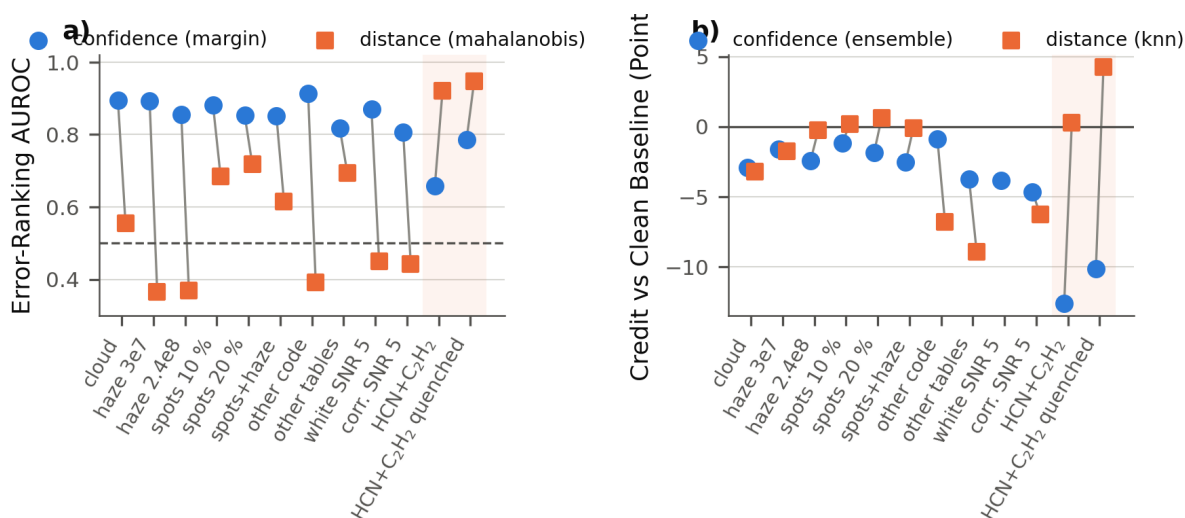


Figure 3. Confidence against distance. (a) The AUROC with which the probability margin (circles) and the Mahalanobis distance (squares) ranked the clean-trained screen's errors, for each mismatch; 0.5 is chance. (b) The credit of the ensemble-disagreement and k-nearest-neighbour decline rules on the randomized screen — accepted-set accuracy minus the clean selective baseline at the same coverage — for the same mismatches. The shaded columns are the two omitted-species cases, on which the ordering of the two instruments inverted.

The consortium's molecule-presence screen gave the refinement that makes the result precise. With HCN and C_2H_2 added at 10^{-7} – 10^{-4} to its test population, its CH_4 , H_2O , CO_2 and NH_3

classifiers lost 0.2–1.1 points (Appendix A); the pre-registered expectation of at least five points on CH₄ was wrong. The two species carry no information about whether methane exceeds 10⁻⁴, so a screen for methane ignored them; they carry the information that defines carbon-rich, so a screen for carbon-rich was broken by their absence. Omitted physics was fatal when it sat on the feature that defined the class being screened — and the species a generic training grid leaves out are exactly the ones that define a chemical regime.

3.3 Tier 1, where triage would happen

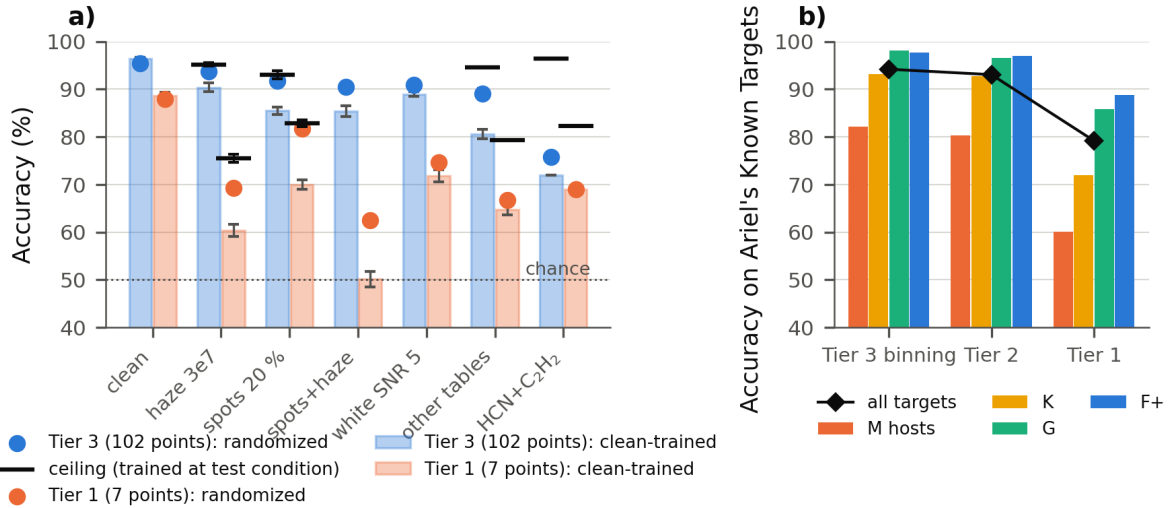


Figure 4. The same mismatches at the two binnings. (a) Accuracy of the clean-trained screen (bars), the randomized screen (circles) and the ceiling from a screen trained at the test condition (ticks) at Tier 3 (102 points, blue) and Tier 1 (7 points, orange); error bars are the standard deviation across the five test sets. (b) The screen on the 965 known planets of Ariel's Mission Candidate Sample under the mission's own noise definition, by tier and host type.

Everything in Section 3.1 is a Tier-3 statement. Triage would happen at Tier 1, on seven numbers per planet, so the map was computed again there (Table 3, Figure 4a) with a screen of the same design trained on the seven-point spectra: 88.7 % on clean spectra, 84.1 % at the tier's SNR of seven.

Table 3. The map at Tier 1 (seven points). Columns as in Table 2; the clean-trained Tier-1 screen scored 88.7 % on clean spectra and the randomized one 87.9 %. Accepted / coverage is for the best decline rule at a threshold that declined 10 % of clean planets; credit is against the clean selective baseline at that coverage.

Mismatch	Clean-trained	Randomized	Ceiling	Best rule: accepted / coverage	Credit
Haze, $3 \times 10^7 \text{ m}^{-3}$	60.3	69.3	75.5	70.7 / 87 %	-21
Star spots, 20 %	70.0	81.7	82.9	84.3 / 88 %	-7
Spots 20 % + haze	50.1	62.5	—	64.9 / 83 %	-28
White noise, SNR 5	71.8	74.6	79.1	—	—
Correlated noise, SNR 5	88.1	86.5	88.1	89.6 / 90 %	-1
Other opacity tables	64.8	66.8	79.3	67.5 / 96 %	-22
HCN + C ₂ H ₂ omitted	66.3	69.0	82.3	71.0 / 86 %	-21

Mismatch	Clean-trained	Randomized	Ceiling	Best rule: accepted / coverage	Credit
Quenched chemistry	92.5	94.2	—	—	gain

Three things changed. First, the losses were larger: a haze that cost six points at Tier 3 cost 28 at Tier 1, because at seven points it removed most of the shape the screen read, and spots and haze together put the screen at chance. Second, the ceilings were low: a screen trained at the test condition recovered only part of each loss — haze to 75 %, the other opacity tables to 79 %, the omitted species to 82 %, white noise to 79 % — so most of every serious loss was not in the data at this binning and no training could recover it. Read against those ceilings the randomized screen did well, reaching 91 % of the ceiling for spots and 60 % for haze, but the ceilings are the story. Third, no decline rule worked: the error-ranking AUROC of every rule on every serious mismatch was 0.57–0.70 and the credits were –7 to –28 points; with seven numbers per planet there was not enough redundancy for either confidence or distance to separate right from wrong. Correlated noise, by contrast, was nearly free at Tier 1, because over seven wide bins it acted like a common offset that normalisation removed, whereas white noise cost 17 points.

The mission's real targets sharpened the picture (Figure 4b). Of the 965 known planets in the Mission Candidate Sample (3), 57 % lay inside the training grid's bulk-parameter box; 29 % exceeded its 300 $M_{\oplus}$ mass cap, 12 % orbited stars larger than 1.7 $R_{\odot}$ and 9 % were cooler than 500 K. Rendered with their real hosts, radii, masses and temperatures, C/O drawn as in the grid, and noise set by the mission's own Tier-2 definition (SNR 7 on the five-scale-height modulation after the catalogue's number of transits, with the ExoSim 2 shape), the clean-trained screen scored 94.2 % at Tier-3 binning, 93.1 % at Tier 2 and 79.2 % at Tier 1, where M-dwarf hosts sat at 60 %. The M-dwarf deficit appeared with no spots at all: it was the cool, small planets around M dwarfs, in the 500–1,000 K band where the label is hardest, not the star. Planets outside the box scored no worse than those inside, so the box edge was not where the screen broke. The achieved signal-to-noise on the planets' real amplitudes under this convention was 15.6 at the median, so the grid's SNR-15 convention was close to Tier-2 reality for this population. The distance-based decline rules, fixed on grid data, would have declined 27–30 % of the real targets under the mission's noise definition against 9–11 % under the grid's: they responded to the noise convention, which is the behaviour Section 3.2 relies on and a warning for deploying them across simulators.

3.4 What a mission should do

The map converts into instructions.

Audit the species list against the chemistry of the class being screened for. A screen for a chemical regime needs the molecules that define that regime in its training grid; the standard five-gas grid lacks the two that define carbon-rich atmospheres, and adding them costs nothing but two opacity tables.

Randomize what is uncertain, and keep a separate alarm for what was not modelled. One randomized grid halved the loss from clouds, haze, spots and their compounds for a one-point cost and reached a third to four-fifths of what single-axis training reached; it also blinded the novelty alarm to whatever it contained, so the alarm should be a second, deliberately fragile model. Confidence-based declining suits distortions of modelled physics and distance-based declining suits novel physics; neither restored clean-level reliability.

Treat the opacity database as the largest reducible risk and noise as the largest irreducible one. Sixteen points separated two opacity databases, 88 % of them recoverable only by training on the alternative; four to five of the eight to twelve points lost at SNR 5 were recoverable by nothing but observing time.

Do not expect Tier-3 reliability at Tier 1. At seven points, randomization still helped, declining did not, and most of every serious loss was unrecoverable by any training. A screen that ranks Tier-1 spectra should be reported at Tier-1 binning, against ceilings, and on the mission's target list.

Run the workflow before a screen enters a ranking. The scripts that produced every number here take any screen and any forward model; the requirements are the re-rendered mismatch sets, an oracle per axis, one randomized grid, the decline rules with clean-fixed thresholds, and the two binnings.

3.5 Limitations and what was not found

The map is conditional on its axis set. Every mismatch was generated by the same family of simulators, so an ingredient outside the set — three-dimensional temperature structure, photochemistry, a wrong host-star model — was not priced; holding axes out narrows this and real spectra would test it, but nothing removes it. Abundances were constant with altitude, one quench scheme was used, the spot contrast was fixed at 0.85, and the noise convention was relative to each planet's amplitude except in the target-list test (Section 2.5). The consortium screen was re-implemented from its text, not its code, and reproduced its Table 6 only in part; its authors had not yet been consulted. The randomized screen had one draw of its ingredient assignments, so its error bars are those of the five test sets rather than of the grid.

Four pre-registered expectations were not met and are reported as such: the randomized grid reached 31–83 % of the single-axis ceilings, not ≥ 90 %; compound mismatches were sub-additive, not super-additive; the omitted species cost the molecule-presence screen under a point, not five; and the divergence of the two noise simulators' shapes on cool hosts cost nothing. A rule from the author's earlier work — that what augmentation can repair is set by how many values a mismatch draws per spectrum — did not survive measurement against the oracle ceilings: augmentation recovered essentially all recoverable loss on every axis, and the apparent rule was an artefact of dividing by clean accuracy. It is withdrawn. The inversion of confidence and distance on novel physics reproduces a known result of the failure-detection literature (35) rather than adding to it, and the M-dwarf concentration of contamination loss is expected physics (21).

4. Conclusion

A workflow for mapping when a simulator-trained screen can be trusted — measuring, for each mismatch between simulator and deployment, the loss, its irreducible part, what a randomized training grid absorbed, whether a decline rule flagged the errors, and what removed the loss — was assembled from standard tools and demonstrated on the consortium's Tier-1 design, re-implemented from its publication, and on a carbon-rich screen built from scratch. Modelled mismatch was largely absorbed and its residual was irreducible information loss; disequilibrium chemistry raised the carbon-rich screen's accuracy; and physics the simulator omitted — HCN and C₂H₂, present in carbon-rich atmospheres and absent from the field's standard training grid — put the carbon-rich screen at chance on carbon-rich planets with confident probabilities, was absorbed by nothing, was detected only by distance-based novelty scores, and was repaired only by adding the species to the forward model, while leaving a molecule-presence screen unchanged. At the Tier-1 binning at which triage would

occur, randomization still helped, no decline rule worked, and most of every serious loss was unrecoverable by any training. Within the tested axes, a screen validated only on the simulator that built it had not been validated for deployment; the workflow released here makes the validation that is possible before launch reproducible on any screen.

Acknowledgements

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Data and Code Availability

Every number in this note is read from a committed result file. The code, the result tables, the pre-registered expectations and their outcomes, the figures, and the documented changes to the forward model (a seeding fix and an aerosol feature in the MultiREx fork, and the added opacity tables) are at <https://github.com/oy2017/BioSignatureDetectionModel>, directory v3; the reliability-map workflow is the sequence in v3/rerun_all.sh. Spectra (5 GB) are regenerated by the scripts in about ten hours; the ExoMolOP tables, Exo-Transmit, FastChem, the PHOENIX atlas and ExoSim 2 are public.

Appendix A. The consortium screen: recipe, deviations, reproduction

The re-implementation followed reference (6) §II.2 and §II.5; neither the Alfnor wrapper nor the trained classifiers were released, so the description in the paper was the only source. Deviations: MultiREx/TauREx 3 with Exo-Transmit tables in place of TauREx 3 with ExoMol k-tables; the ExoSim 2 noise shape scaled to the Tier-1 requirement per target in place of ArielRad's Tier-1 noise; the 965 known planets of the 2026 candidate list in place of their 1,000 (which included TESS predictions); training spectra noised once rather than resampled. Reproduction at the 10^{-4} threshold (the published Table 6 in parentheses): CH₄ 70–72 (82–87), H₂O 63–69 (71–78), CO₂ 60 (79–83), NH₃ 76–78 (82–87) %. Checks: moving the AIRS-CH₀ split from 2.76 to 3.0 or 2.5 μm changed the four by at most 2 points; correcting the noise for the integer number of Tier-1 transits per target (a target needing N transits reaches the requirement with $N - 0.5$ on average) changed them by at most 1.4; scaling the noise to 0.75, 0.5, 0.35 and 0.25 of the requirement gave CH₄ 76, 78, 82, 86, H₂O 70, 72, 72, 86 and NH₃ 80, 82, 84, 61 %, so three of the four classifiers reached their published values at roughly a third of the re-implementation's noise, while CO₂ stayed at the majority rate (60–62 %) until the noise was a quarter (71 %). Mismatch costs on the re-implemented screen at the requirement noise, mean over the four classifiers: haze $3 \times 10^7 \text{ m}^{-3}$ –8 to –17, spots 20 % –6 to –13, cloud deck at 10^2 Pa –6 to –16, noise doubled –5 to –8, HCN + C₂H₂ at 10^{-7} – 10^{-4} –0.2 to –1.1 points. Decline rules on the re-implemented screen: ensemble disagreement across the four classifiers kept ~100 % of planets under haze (all four agreed on the wrong answer), while the

k-nearest-neighbour distance declined 97 % — the same confidence-versus-distance split as on the carbon-rich screen, with the distance rule acting as a population alarm rather than a per-planet fix. Table 4 gives every number.

Table 4. The consortium's Tier-1 screen, re-implemented: accuracy (%) at the 10^{-4} abundance threshold for each molecule and classifier on the clean test population and under each mismatch, with the published range from reference (6) Table 6 for comparison. The "vote" rows are the majority vote of the four classifiers after the ensemble-disagreement or k-nearest-neighbour decline rule, on the accepted planets only.

Molecule	Classifier	Their Table 6	Clean	Haze 3×10^7	Cloud 10^2 Pa	HCN + C_2H_2	Spots 10 %	Spots 20 %	Noise $\times 2$	Noise $\times 3$
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Appendix B. Full envelope tables

Tables 5 and 6 give, for every mismatch and every decline rule at each binning, the accuracy of all planets and, per rule, the accepted-set accuracy with the coverage kept and the credit against the clean selective baseline at that coverage (points). Table 7 gives the clean-trained, randomized, held-out and ceiling accuracies at both binnings. The source files are ariel_trust_envelope, tier1_trust_envelope, ariel_trust_randomized, tier1_trust_randomized, ariel_oracle and tier1_oracle in v3/results.

Table 5. The envelope at Tier 3 (102 points), randomized screen. Cells: accepted-set accuracy % (coverage %; credit in points). Thresholds declined 10 % of clean planets.

Mismatch	All	Ensemble	Margin	Mahalanobis	k-NN
Clean	95.4	96.7 (90 %; +0.0)	98.4 (90 %; +0.0)	95.2 (90 %; +0.0)	94.9 (90 %; +0.0)
Cloud deck, 10^5 Pa	95.3	96.7 (90 %; -0.0)	98.2 (90 %; -0.2)	94.8 (85 %; -0.2)	94.7 (88 %; -0.2)
Cloud deck, 10^4 Pa	94.5	96.5 (87 %; -0.6)	97.9 (89 %; -0.6)	93.4 (63 %; -1.2)	93.4 (70 %; -0.5)
Cloud deck, 10^3 Pa	92.7	95.8 (76 %; -2.9)	97.4 (85 %; -1.4)	91.7 (25 %; -4.7)	92.2 (39 %; -3.2)
Cloud deck, 10^2 Pa (out of range)	86.7	92.8 (57 %; -6.9)	93.9 (77 %; -5.5)	89.5 (2 %; -6.3)	88.5 (5 %; -9.8)
Cloud deck, 10 Pa (out of range)	65.2	72.4 (21 %; -27.6)	73.0 (55 %; -26.9)	100.0 (0 %; +0.0)	90.9 (0 %; -9.1)
Haze, 2×10^5 m $^{-3}$	95.4	96.6 (90 %; -0.0)	98.3 (90 %; -0.1)	95.0 (89 %; -0.1)	94.9 (89 %; -0.0)
Haze, 2×10^6 m $^{-3}$	94.8	96.2 (91 %; -0.4)	98.0 (90 %; -0.4)	94.1 (82 %; -0.8)	94.0 (85 %; -0.7)
Haze, 3×10^7 m $^{-3}$	93.7	95.1 (90 %; -1.6)	97.4 (89 %; -1.2)	92.7 (70 %; -2.0)	92.3 (72 %; -1.7)
Haze, 2.4×10^8 m $^{-3}$	93.1	95.0 (85 %; -2.4)	97.1 (87 %; -1.6)	93.7 (40 %; -1.1)	93.6 (49 %; -0.3)
Haze, 10^{10} m $^{-3}$ (out of range)	70.1	74.2 (31 %; -25.8)	82.2 (37 %; -17.8)	91.5 (4 %; -5.2)	91.7 (7 %; -6.4)
Star spots, 2 %	94.4	96.2 (90 %; -0.5)	98.0 (90 %; -0.4)	94.2 (86 %; -0.9)	94.1 (86 %; -0.7)
Star spots, 5 %	93.8	96.0 (89 %; -0.8)	97.6 (89 %; -0.9)	93.9 (82 %; -1.1)	94.1 (79 %; -0.3)
Star spots, 10 %	92.9	96.0 (87 %; -1.2)	97.3 (87 %; -1.4)	92.9 (80 %; -1.9)	94.2 (71 %; +0.2)
Star spots, 20 %	91.7	96.1 (82 %; -1.9)	97.2 (84 %; -1.7)	91.2 (82 %; -3.8)	94.2 (63 %; +0.6)
Spots + faculae (out of range)	93.0	95.9 (87 %; -1.2)	97.2 (88 %; -1.4)	93.3 (77 %; -1.4)	94.1 (71 %; +0.1)
Faculae, 10 % (out of range)	93.4	97.5 (83 %; -0.3)	96.6 (90 %; -1.8)	93.1 (91 %; -2.1)	93.2 (86 %; -1.6)

Mismatch	All	Ensemble	Margin	Mahalanobis	k-NN
Quenched chemistry	98.2	98.5 (99 %; +2.8)	98.7 (99 %; +2.7)	98.0 (80 %; +3.1)	98.0 (83 %; +3.4)
Other radiative-transfer code	90.2	96.8 (84 %; -0.9)	94.0 (93 %; -3.8)	87.8 (57 %; -6.9)	87.0 (67 %; -6.8)
Other opacity tables	89.1	96.0 (52 %; -3.8)	95.9 (78 %; -3.4)	97.4 (4 %; +0.8)	88.8 (21 %; -8.9)
Spots 10 % + haze 3×10^7	92.1	95.6 (86 %; -1.8)	97.0 (86 %; -1.8)	91.3 (81 %; -3.6)	93.9 (59 %; +0.3)
Spots 20 % + haze 3×10^7	90.5	95.5 (81 %; -2.5)	96.3 (83 %; -2.6)	90.1 (89 %; -5.0)	93.7 (54 %; -0.1)
HCN + C ₂ H ₂ omitted	75.8	86.6 (71 %; -12.6)	77.7 (86 %; -21.1)	94.1 (62 %; -0.5)	94.0 (66 %; +0.3)
HCN + C ₂ H ₂ omitted, quenched	75.7	88.9 (73 %; -10.2)	77.5 (92 %; -20.5)	98.5 (43 %; +3.7)	98.1 (53 %; +4.3)
White noise, SNR 12	94.4	96.7 (87 %; -0.4)	97.7 (89 %; -0.8)	93.5 (45 %; -1.3)	92.8 (75 %; -1.3)
White noise, SNR 8	92.5	96.1 (80 %; -2.1)	96.9 (88 %; -1.7)	100.0 (0 %; +0.0)	95.4 (23 %; -2.0)
White noise, SNR 5	90.8	95.8 (60 %; -3.9)	95.9 (85 %; -2.9)	declines all	declines all
Correlated noise, SNR 12	93.7	96.0 (88 %; -1.0)	97.2 (90 %; -1.2)	92.6 (77 %; -2.2)	92.0 (77 %; -2.3)
Correlated noise, SNR 8	91.1	95.9 (80 %; -2.4)	95.0 (90 %; -3.5)	92.0 (34 %; -3.3)	91.6 (34 %; -4.4)
Correlated noise, SNR 5	87.2	94.9 (65 %; -4.7)	91.2 (88 %; -7.4)	89.3 (6 %; -7.8)	91.6 (4 %; -6.3)

Table 6. The envelope at Tier 1 (7 points), randomized screen. Cells as in Table 5.

Mismatch	All	Ensemble	Margin	Mahalanobis	k-NN
Clean	87.9	91.0 (90 %; +0.0)	91.2 (90 %; +0.0)	87.7 (90 %; +0.0)	88.2 (90 %; +0.0)
Cloud deck, 10^5 Pa	86.7	90.0 (89 %; -1.3)	90.4 (88 %; -1.5)	86.5 (88 %; -1.2)	87.4 (88 %; -0.8)
Cloud deck, 10^4 Pa	81.8	85.7 (86 %; -6.4)	86.5 (85 %; -6.5)	82.9 (77 %; -4.3)	84.5 (78 %; -4.0)
Cloud deck, 10^3 Pa	75.6	79.0 (81 %; -14.0)	79.9 (82 %; -14.0)	78.3 (55 %; -8.0)	80.0 (57 %; -9.3)
Cloud deck, 10^2 Pa (out of range)	64.5	67.8 (72 %; -27.1)	68.9 (74 %; -26.7)	71.6 (27 %; -12.2)	73.2 (27 %; -18.5)
Cloud deck, 10 Pa (out of range)	52.7	53.1 (60 %; -43.0)	53.8 (65 %; -43.0)	56.8 (6 %; -18.6)	55.4 (6 %; -40.5)
Haze, 2×10^5 m ⁻³	87.7	90.8 (90 %; -0.1)	91.3 (90 %; +0.1)	87.5 (91 %; -0.2)	88.0 (89 %; -0.4)
Haze, 2×10^6 m ⁻³	87.4	90.0 (91 %; -0.7)	90.9 (88 %; -1.0)	86.3 (84 %; -1.3)	87.0 (77 %; -1.5)
Haze, 3×10^7 m ⁻³	69.3	70.7 (87 %; -21.0)	75.3 (66 %; -21.4)	67.9 (73 %; -19.2)	67.3 (64 %; -21.9)
Haze, 2.4×10^8 m ⁻³	60.5	61.3 (71 %; -33.6)	65.9 (47 %; -32.4)	58.0 (41 %; -27.5)	60.4 (60 %; -28.9)
Haze, 10^{10} m ⁻³ (out of range)	50.2	51.6 (47 %; -45.8)	51.5 (46 %; -46.8)	61.5 (2 %; -8.9)	52.3 (10 %; -42.6)
Star spots, 2 %	87.8	90.2 (91 %; -0.5)	91.1 (90 %; +0.0)	87.9 (85 %; +0.3)	88.8 (83 %; +0.4)
Star spots, 5 %	86.5	89.0 (91 %; -1.8)	90.8 (87 %; -1.4)	86.5 (79 %; -0.9)	87.5 (76 %; -1.0)
Star spots, 10 %	84.9	87.1 (90 %; -4.0)	90.7 (80 %; -3.5)	84.2 (76 %; -3.0)	84.7 (73 %; -3.9)
Star spots, 20 %	81.7	84.3 (87 %; -7.2)	91.1 (68 %; -5.5)	79.4 (79 %; -7.9)	79.5 (77 %; -9.0)
Spots + faculae (out of range)	85.1	87.4 (90 %; -3.5)	90.6 (82 %; -3.1)	84.8 (74 %; -2.4)	85.6 (72 %; -3.1)

Mismatch	All	Ensemble	Margin	Mahalanobis	k-NN
Faculae, 10 % (out of range)	84.6	87.4 (89 %; -3.9)	87.8 (90 %; -3.3)	85.9 (83 %; -1.7)	86.1 (89 %; -2.2)
Quenched chemistry	94.2	95.3 (96 %; +6.0)	95.5 (96 %; +6.3)	94.8 (82 %; +7.2)	95.9 (80 %; +7.6)
Other radiative-transfer code	86.8	89.9 (90 %; -1.0)	90.1 (90 %; -0.9)	86.6 (84 %; -1.0)	87.1 (83 %; -1.2)
Other opacity tables	66.8	67.5 (96 %; -21.8)	67.4 (96 %; -21.9)	68.2 (85 %; -19.4)	68.0 (87 %; -20.3)
Spots 10 % + haze 3×10^7	65.4	67.2 (86 %; -24.7)	75.3 (46 %; -22.9)	65.1 (92 %; -22.7)	64.5 (89 %; -23.8)
Spots 20 % + haze 3×10^7	62.5	64.9 (83 %; -27.8)	76.2 (33 %; -23.0)	62.4 (97 %; -25.5)	62.1 (96 %; -26.0)
HCN + C ₂ H ₂ omitted	69.0	71.0 (86 %; -21.1)	71.8 (85 %; -21.1)	67.1 (91 %; -20.6)	68.4 (91 %; -19.9)
HCN + C ₂ H ₂ omitted, quenched	70.6	72.1 (90 %; -18.8)	72.7 (89 %; -18.8)	67.0 (84 %; -20.6)	68.9 (82 %; -19.5)
White noise, SNR 12	86.7	89.5 (90 %; -1.4)	89.8 (91 %; -1.2)	86.3 (86 %; -1.3)	87.3 (86 %; -1.1)
White noise, SNR 8	83.1	85.9 (88 %; -5.4)	86.1 (90 %; -5.1)	82.0 (69 %; -5.0)	84.1 (67 %; -4.7)
White noise, SNR 5	74.6	78.1 (82 %; -14.8)	77.1 (88 %; -14.8)	73.2 (42 %; -12.4)	78.1 (39 %; -12.6)
Correlated noise, SNR 12	87.6	91.0 (90 %; +0.1)	91.3 (90 %; +0.1)	87.5 (89 %; -0.2)	88.0 (89 %; -0.3)
Correlated noise, SNR 8	87.3	90.1 (90 %; -0.8)	90.5 (91 %; -0.5)	86.9 (87 %; -0.8)	87.8 (86 %; -0.5)
Correlated noise, SNR 5	86.5	89.6 (90 %; -1.4)	89.8 (90 %; -1.4)	86.2 (80 %; -1.3)	86.6 (78 %; -1.9)

Table 7. Clean-trained, randomized, held-out and ceiling accuracies (%) at both binnings. Held-out is the randomized grid rebuilt without the case's ingredient (for the two codes and the omitted species every grid is held out and the randomized value is repeated); ceiling is a screen trained at the test condition.

Mismatch	Tier 3: clean-trained	Tier 3: randomized	Tier 3: held-out	Tier 3: ceiling	Tier 1: clean-trained	Tier 1: randomized	Tier 1: ceiling
Clean	96.5	95.4	—	—	88.7	87.9	—
Cloud deck, 10^5 Pa	96.4	95.3	95.3	—	87.8	86.7	—
Cloud deck, 10^4 Pa	95.2	94.5	94.1	—	83.8	81.8	—
Cloud deck, 10^3 Pa	92.4	92.7	92.2	—	76.5	75.6	—
Cloud deck, 10^2 Pa (out of range)	85.5	86.7	85.8	—	64.9	64.5	—
Cloud deck, 10 Pa (out of range)	63.7	65.2	64.5	—	52.2	52.7	—
Haze, 2×10^5 m ⁻³	96.3	95.4	95.5	—	88.6	87.7	—

Mismatch	Tier 3: clean-trained	Tier 3: randomized	Tier 3: held-out	Tier 3: ceiling	Tier 1: clean-trained	Tier 1: randomized	Tier 1: ceiling
Haze, 2×10^6 m ⁻³	94.7	94.8	94.6	—	87.3	87.4	—
Haze, 3×10^7 m ⁻³	90.4	93.7	92.7	95.2	60.3	69.3	75.5
Haze, 2.4×10^8 m ⁻³	88.1	93.1	91.5	—	50.2	60.5	—
Haze, 10^{10} m ⁻³ (out of range)	61.3	70.1	66.2	—	52.6	50.2	—
Star spots, 2 %	94.0	94.4	94.1	—	87.4	87.8	—
Star spots, 5 %	91.5	93.8	92.3	—	84.1	86.5	—
Star spots, 10 %	88.8	92.9	89.8	94.2	78.3	84.9	85.5
Star spots, 20 %	85.5	91.7	86.5	93.0	70.0	81.7	82.9
Spots + faculae (out of range)	89.2	93.0	90.0	—	79.8	85.1	—
Faculae, 10 % (out of range)	93.0	93.4	93.3	—	86.9	84.6	—
Quenched chemistry	98.3	98.2	98.1	—	92.5	94.2	—
Other radiative-transfer code	90.0	90.2	90.2	95.5	86.8	86.8	87.5
Other opacity tables	80.6	89.1	89.1	94.6	64.8	66.8	79.3
Spots 10 % + haze 3×10^7	87.6	92.1	—	—	50.7	65.4	—
Spots 20 % + haze 3×10^7	85.4	90.5	—	—	50.1	62.5	—
HCN + C ₂ H ₂ omitted	75.8	75.8	—	96.4	69.0	69.0	82.3
HCN + C ₂ H ₂ omitted, quenched	75.7	75.7	—	—	70.6	70.6	—
White noise, SNR 12	94.5	94.4	94.0	—	86.9	86.7	—
White noise, SNR 8	92.2	92.5	92.1	—	81.0	83.1	—
White noise, SNR 5	88.9	90.8	89.9	—	71.8	74.6	—
Correlated noise, SNR 12	93.0	93.7	93.6	—	88.7	87.6	—
Correlated noise, SNR 8	89.2	91.1	90.7	—	88.8	87.3	—
Correlated noise, SNR 5	84.1	87.2	85.4	—	88.1	86.5	—

Appendix C. Conformal prediction and calibration under mismatch

Split-conformal sets calibrated on clean test planets to 90 % coverage realised 90.8 % on clean spectra and 79 (cloud 10^3 Pa), 88 (haze 3×10^7), 81 (spots 20 %), 63 (other opacity tables), 64 and 62 (omitted species, equilibrium and quenched), 77 and 75 % (white and correlated noise at SNR 5); in every case the shortfall appeared as empty sets, not ambiguous ones. The clean-trained screen's expected calibration error was 0.009 on clean spectra, 0.08–0.11 under haze and spots, 0.17 under the other opacity tables and 0.25–0.33 under the omitted species, where the mean probability assigned to a true carbon-rich planet fell from 0.96 to 0.46 and 0.34.

Appendix D. Forward model and its audit

NH₃ carried no absorption in the forward model of the author's earlier work (MultiREx ships no NH₃ table), a documented simplification that this note's thesis made indefensible; the Exo-Transmit NH₃ table was added and every result regenerated (haze and opacity-table costs rose; the contamination host gap narrowed; every qualitative statement survived). The MultiREx fork used here fixes a seeding defect in the released package — objects created within the same second draw identical parameters and shuffled clones reproduce their parent — which the grids of this note never exercised because parameters were drawn externally, and adds the cloud and Mie-haze contributions. The quench prescription, the eddy-diffusion sweep and the chemistry checks are documented in the repository (v3/TRUST_IDEA.md).

Appendix E. Pre-registered expectations and outcomes

Committed before the runs: randomized grid within one point of clean (met: 1.06) and ≥ 90 % of ceilings (not met: 31–83 %); compounds super-additive (not met); probability margin the worst detector under re-rendered shifts (not met — it was the best on distorted physics and the worst on novel physics); distance scores detect re-rendered physics (met for detection, wrong about usefulness except on novel physics); the absorb/detect trade-off (met); omitted species cost the consortium screen ≥ 5 points on CH₄ (not met: ≤ 1.1); the Tier-1 consortium screen reproduced within ± 5 points (met for NH₃ and CH₄ at reduced noise only).

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